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Computer Architecture - Assignment 1

Q1) P2 and P1 are ^{given} assumed to have same no. of instruction as the
so Data

P2

P1

$$N_{inst} = IC$$

$$N_{inst} = IC$$

$$CPI_2 = 1.2 CPI_1$$

$$CPI_1 = CPI_1$$

$$f_2 = x$$

$$f_1 = 1 \text{ GHz}$$

$$T_2 = 0.75 T_1$$

$$T_1 = T_1$$

$$\frac{T_2}{T_1} = \frac{CPI_1 (1.2) \times IC}{f_2} \div \frac{CPI_1 \times IC}{f_1}$$

$$0.75 = \frac{1.2 \times f_1}{f_2}$$

$$f_2 = \frac{1.2 \times 1 \text{ (GHz)}}{0.75}$$

$$f_2 = 1.6 \text{ GHz} \quad \text{Answer}$$

Q2) Data

$$MTTF \text{ of 1 drive} = 2000000$$

$$\text{Server} = 25000 \quad \left\{ \text{each server has 4 drive} \right.$$

$$\text{Total Drive in warehouse} = 4(25000) = 100000$$

$$a) FIT = \frac{1}{MTTF} \times 10^9 = \frac{1}{2000000} = 500 \text{ FIT}$$

$$b) \lambda_{sys} = \text{Total Drive} \times \frac{1}{MTTF_{drive}} = 100000 \times \frac{1}{2000000} = \frac{1}{20} = 0.05 \text{ failure/hr}$$

$$c) \text{MTTF}_{\text{sys}} = \frac{1}{\lambda_{\text{sys}}} = \frac{1}{0.05} = 20 \text{ hours}$$

$$d) \lambda_{\text{new-sys}} = \frac{100000}{1500000} = \frac{1}{15} \text{ failure/hr}$$

$$\text{MTTF}_{\text{New sys}} = \frac{1}{\lambda_{\text{New-sys}}} = \frac{1}{(1/15)} = 15 \text{ hours}$$

$$e) \text{Reliability} = e^{-\lambda t}$$

Note:- Reducing MTTF from 2000000 to 1500000 hrs is not an improvement in reliability. It is a degradation

$$\frac{\text{Ratio original WSC MTTF}}{\text{Ratio new WSC MTTF}} = \frac{20}{15} = 1.33$$

$\therefore$ The original system was $1.33 \times$ more reliable in terms of MTTF than the reduced MTTF system

Q3a) Overall CPI of Program = ?

$$\text{CPI}_{\text{effective}} = (0.4)(1) + (0.25)(3) + (0.25)(3) + (0.1)(\overset{10}{\cancel{10}}) = 2.9$$

$$3b) \text{CPU}_{\text{Time}} = \frac{\text{Instruction Count} \times \text{CPI}_{\text{effective}}}{\text{Clock Speed}}$$

$$= \frac{(30 \times 10^9) \times 2.9}{3 \times 10^9}$$

$$= 29 \text{ seconds}$$

Q4a) The 25% year era was driven almost entirely by one lever: semiconductor technology scaling (Moore's Law), smaller, faster transistors on each new process node, with relatively simple, largely sequential (non-pipelined or lightly pipelined) processor designs. That's a single multiplicative source of gain.

The 52% year era added a second, independent lever on top of the first. Architectural exploitation of Instruction Level Parallelism (ILP). This is the RISC era pipelining, superscalar, parallelism, cache etc. All let a single core complete more instructions per clock cycle, not just run the clock faster. Since technology scaling (more/faster transistors) and microarchitectural cleverness, are roughly multiplicative, combining them produced a much steeper curve than either alone.

4b) A few things converged.

- Diminishing returns from ILP: after two decades of deeper pipelines and wider issue widths, most real programs simply don't contain enough independent, parallel instructions to keep feeding an even wider superscalar core. "ILP wall"
- Memory Wall: Memory access latency improved far more slowly than processor speed, so processors increasingly stalled waiting on data regardless of how fast the core itself could run.
- Growing design/verification complexity and power cost of ever more aggressive out of order/speculative hardware, for shrinking marginal gains.

Industry response was to instead of making one core do more per cycle, put multiple cores on a chip and rely on Task-Level Parallelism. This is exactly why the chart's growth rate drops in stages as the industry shifts to multi core total power consumption and heat generation increase when they are active. Since available power and cooling system could no longer keep increasing clock frequency as rapidly

Date: _____

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as before. Therefore performance growth slowed as the industry shifted from faster single cores to more cores at relatively flatter frequencies.

4c) Dennard Scaling (1974) observed that as transistors shrink by a factor, you could scale voltage down by roughly the same factor, keeping power density roughly constant. So each new, smaller generation of transistors could run at a higher clock frequency without the chip getting hotter.

Around 2004-2006, Dennard Scaling broke down. For voltage could no longer ^{be lowered} ~~be~~ further without leakage current growing exponentially. Without voltage scaling down in step with transistor size, power density started rising with each new generation instead of staying flat.

This directly produced the Power Wall: $P = \frac{1}{2} CV^2 F$. With V stuck at a _{dynamic}

floor any further increase in clock frequency translates almost directly into more power and more heat. Architects hit a hard thermal ceiling on how fast they could clock a single core.

4d) Since Dennard Scaling no longer lets voltage shrink with each process generation the power dynamic relationship means frequency increases now cost power almost linearly. Beyond a certain clock speed, the chip simply generates more heat than it can safely dissipate. (we would require cooling solutions that are impractical for main products.

Meanwhile the ILP wall means that even if we could clock higher, we wouldn't extract proportionally more useful work per cycle anyway.

Hence due to all this the industry move to multicore, specialized accelerators. (GPU and such) rather than continuing to chase raw clock frequency.

Q5a) Standardized Benchmarks are needed to compare the performance of different processors fairly. Different processors may have different architectures, clock speeds, cache size and strengths. By running the same standardized programs on each system, their performance can be measured using a common basis instead of relying only on specifications such as clock frequency.

5b) SPECint measures processor performance for integer based workloads, which are common in general purpose computing.

SPECfp measures performance for floating point workloads, which are common in scientific, engineering and numerical applications. Both are used to evaluate different processor performance. SPECint and SPECfp are the two halves of the SPEC CPU benchmark suite.

5c) SPEC provides a standardized set of programs and rules for testing different computer systems. The same workloads are run on each system and their execution performance is measured using the same methodology. This makes the results more consistent and allows processors from different manufacturers and architectures to be compared fairly.

5d) A single program only tests a limited type of workload and may not represent the processor's overall performance. A suite contains different programs that exercise different parts and capabilities of the processor. Therefore, using multiple benchmarks provides a more representative measure of performance.

5e) A higher SPEC performance score indicates that the processor performs the benchmark workloads faster than a system with a lower score. In general, a higher score means better performance for the types of applications represented

Date: _____

M	T	W	T	F	S	S
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by the benchmark suite. However it does not mean that the processor will be faster for every possible applications.

5f) One limitation of a benchmarks is that they may not accurately represent the actual workload of every user or organization. A processor may perform very well on a particular benchmark suite but perform differently on applications such as data base, web server, or other specialized workloads. Therefore benchmark results should not be considered a complete measure of overall computer performance.